

FIRST BI-MONTHLY ISSUE—

A.N.C.

PLANET

stories

TRADE
MARK
REG



CARGO TO CALLISTO

by JAY DREXEL

Another famous

STAR-MOUSE story

by FREDERIC BROWN



A man and a woman—survivors of chaos—alone in dead-star vastness..

THE LAST TWO ALIVE!

A Novel of Exploding Suns by ALFRED COPPEL

Also ALLEN K. LANG • JOHN D. McDONALD • C. H. LIDDELL

FIRST BI-MONTHLY ISSUE—

A.N.C.

PLANET stories

TRADE
MARK
REG



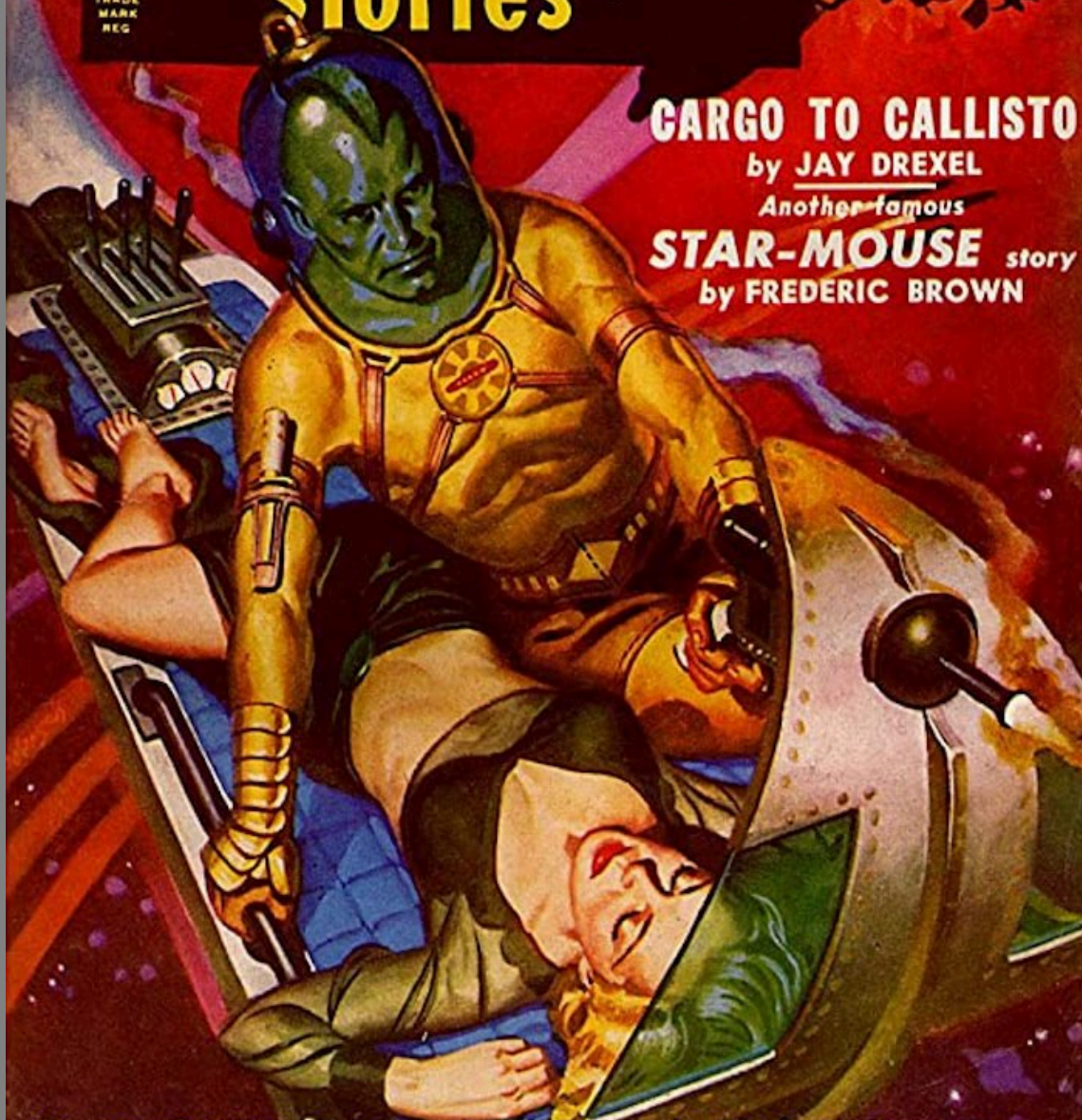
CARGO TO CALLISTO

by JAY DREXEL

Another famous

STAR-MOUSE story

by **FREDERIC BROWN**



A man and a woman—survivors of chaos—alone in dead-star vastness...

THE LAST TWO ALIVE!

A Novel of Exploding Suns by **ALFRED COPPEL**

Also **ALLEN K. LANG** • **JOHN D. McDONALD** • **C. H. LIDDELL**

The Project Gutenberg eBook of Machine of Klamugra

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Machine of Klamugra

Author: Allen Kim Lang

Illustrator: Earl Mayan

Release date: February 26, 2021 [eBook #64640]

Most recently updated: October 18, 2024

Language: English

Other information and formats: www.gutenberg.org/ebooks/64640

Credits: Greg Weeks, Mary Meehan and the Online Distributed Proofreading Team at <http://www.pgdp.net>

*** START OF THE PROJECT GUTENBERG EBOOK MACHINE OF
KLAMUGRA ***

Machine of KLAMUGRA

By Allen K. Lang

**Captain Barnaby and Lieutenant Teajun stood
at the brink of that vast stone amphitheater,
staring wonderingly down at half-an-acre of
gadget. This glittering mass of million-year
clockwork was the Machine ... and soon it was
to judge them for their crime against Mars!**

**[Transcriber's Note: This etext was produced from
Planet Stories November 1950.**

**Extensive research did not uncover any evidence that
the U.S. copyright on this publication was renewed.]**

Klaggchallak, his fur nose-flaps pulled tight against his nostrils, stumbled up to the gleaming pinnacle of steel that seemed to offer shelter against the night. He felt a dust-storm gathering in the west, and knew that not even the tough skin of a Martian priest could withstand the angry whippings of sand lashed up by the wind-warlocks of the desert.

The old priest drew a tiny, folded *mal*-skin tent from his back-pack. Without haste, for he knew that the elder gods of Mars were watching his safety, Klaggchallak pitched the tent against the west stabilizer of the rocket, drawing the tough hide down to form a floor-flap and fastening it to the steel of the stabilizer with tough *mal*-hoof glue, which would hold fast in the fiercest winds of Mars. He looked for the sun and found it low in the evening sky, then crawled leisurely into the yurt, pulling the door-flap down after him and gluing it to the floor. He had for himself a secure cocoon into which the sand-devils could not force their probing fingers. Before he slept, the old priest fingered his beads, reciting his evening invocation to various benevolent and protective gods.

The falling sun threw a dancing star against the hull of the ship standing tall in its tail-chocks. A bewildering wail, the banshee-call of "Danger; ship jetting off!" sounded; but Klaggchallak slept on, hearing through his dreams only the howling of the wind.

Sixty seconds later, as prescribed in the General Regulations of the Extraterrestrial Service, a second sound began, that most fearful of noises, the sirening of the rocket exhaust. The Martian in his skin tent awakened and felt fear gnaw at his bones; fear induced by subsonic tremors from the rocket blast. Klaggchallak reached for his beads as the heat soaked into his thick, wrinkled skin.

In a moment the floor sagged beneath him. With the *mal*-skin pouch dangling ridiculously from its tail assembly, the EXTS rocket *Vulcan* rose with great gentleness from its tail-chocks, pushed up on its spraying jets.

Four seconds later the ship was a ruby flame above the low hills. Eight seconds later a charred bipod, a bifurcated cinder, tumbled down from space to strike near the jetoff field, where the *Vulcan*'s tail-chocks glowed dull red and the blackened ground smoldered. A moment later a bracelet of blast-

welded beads tumbled down from the sky, falling near the carbon hulk that a few seconds before had been Klaggchallak, a Martian priest great in wisdom and in honor among his people.

Captain Jan Barnaby and Lieutenant Kim Teajun of the Extraterrestrial Service stood before the Board of Inquiry at Denver, D. F. The President of the Board, a Chief Commander's star-on-silver gleaming at his right collar point, opened the proceedings:

"The military rocket *Vulcan*, EXTS light cruiser, is accused by the Martian authorities of causing the death of one Klaggchallak, a priest. They further claim that the death of Klaggchallak was caused by criminal negligence on the part of the pilot and co-pilot of the rocket, Captain Jan Barnaby and Lieutenant Kim Teajun, respectively. Such neglect being within the definition of murder in the Martian legal code, the Judging Authority of Mars demands that we deliver these two men to them for trial and eventual punishment."

The Commander stroked his grey hair thoughtfully as he looked up from his report to the two unhappy officers before him. "At ease, gentlemen." Barnaby and Teajun slumped. "While I'm inclined to agree with you two that Klaggchallak's frying was his own fool fault, I must say that you picked a damned poor time to become the instruments of his immolation. We had hoped to establish an extra-territoriality agreement with Mars, but the death of old Klaggchallak puts that out of the question. To further Martian tranquility, you men will have to return to Mars and face the Judging Authority there. If my feelings were all that is at stake, gentlemen, I'd tell the Marties to go trippingly to hell, and keep you here on Earth. But to do this would mean we'd be forced to abandon our bases and mines and surveys on all Mars. We'd be giving our European competitors a clear field." The Commander folded the report neatly, once and again. "Captain Barnaby and Lieutenant Kim, you'll be on the next Mars-ward ship. We can't help you if you're convicted by our fuzzy friends. You'll have to stay there and take whatever punishment they demand."

Kim was remembering a scene he and Barnaby had witnessed at Klamugra, the seat of the Martian Judging Authority. A Martian, convicted of murder, was being executed atop a high metal platform. A large portion of the city's population was gathered before the platform, watching the edifying spectacle of a fellow-Martian dying with horrifying slowness as the chocks of a vise pressed into his skull. They were bearcats for gladiatorial amusement.

"Do you gentlemen have any questions?"

Lieutenant Kim glanced at Captain Barnaby, then spoke. "Yes, sir. I'd like to know how long we're going to let the Marties push us around this way. Thirteen Martian priests are on our payroll, just because they demand it. We've got to stay five kilometers away from their cities, or pay a five-hundred credit fine. We can't spit without special permission from the Grand Council of Mars. We don't think like they do; why should we submit to being judged by their million-year-old laws? In all respect, sir, why does our Service act so weak?"

The Commander made a pyramid of thumbs and forefingers, and considered it. "Lieutenant Kim, I've been asking myself that question for the last ten years. We've had to pay tribute to gain the Marties' permission to stay on their god-forsaken planet. That tribute represents half the operating expense of the Martian Department of the Service, credits that should be spent on new ships and more men. We've behaved like a bunch of patsies ever since von Munger and Ley landed on Mars.

"Still, we're all soldiers, and we must follow regulations. We mustn't disturb the indigenous population on Mars; that's Regulation 'A-1.' If our policies grow distasteful to the Marties, they may call in the Europeans to take our place. We wouldn't like that. It's bad form to admit it, gentlemen, but I'm ashamed to give you this order. You're to jet off for Mars tomorrow morning; and on arrival at Klamugra, to deliver yourselves over to the Martian Judging Authority." The Commander rapped his gavel and stood; the two officers before the Board snapped to attention. "Board of Inquiry dismissed."

Fully aware that tomorrow's jettison would multiply by eight the hangovers they were breeding, Captain Barnaby and Lieutenant Kim sat that evening in the Denver Dive, alternating drinks of European vodka with rounds of California moon-dew. As Kim said: "Drink as much as you like, Barnaby; we're not driving in the morning."

"Tell me," Barnaby demanded of his co-pilot, "what you're thinking of, you Martie-roasting fiend of a Korean."

"I was considering the memory of the 'shlunk!' that Martian murderer's skull made when it finally gave in, that day at Klamugra. Do you remember, hard-headed Yankee?" Kim's eyes followed the blonde ecdysiast across the stage more from habit than present interest.

"Why did you have to remember that? 'Shlunk!'—ugh!"

"We're going to have to squirm out of this, Barnaby-*sunsang*," Kim said. "We'll have to beat that rap at Klamugra. It's not that I wish to avoid putting my head in a vise; it's only that it hurts me to see the Extraterrestrial Service made a monkey of this way. In a way it will even be a shame if we get off. Think of all the Marties who will miss the opportunity to see your punkin head smashed."

"You orientals have noble souls, Kim."

The blonde stripper, having uncovered as much of herself as she could without resorting to dissection, jumped down from the stage and walked over to the two EXTS officers. "Would you gentlemen like to buy me a drink?" she asked.

Kim's eyes roved abroad in a brief anatomy lesson, but Barnaby said, "I'll buy you one a couple of weeks from now, if I'm not laid up somewhere with a splitting headache." He stood unsteadily and tossed a ten-credit certificate on the table. "If you're really thirsty, get a drink out of that."

Kim reluctantly followed his superior officer from the bar. At the door he turned and called back to the blonde, "Don't catch cold, child. I'll be back."

The dawn jettie was miserable, as jetties always are. Four days brought the ship within falling-distance of Mars; soon the jets thundered as it backed into a pocket of hills outside Klamagra. The air-pumps hammered to bring the air pressure inside the hull gradually down to that of the outside, so that instruments and equipment wouldn't be subjected to a sudden lowering of pressure. The men inside the ship slipped plastic helmets over their heads, checked the tiny air-pumps on their shoulders, and drew on heavy gloves and boots.

When the port swung open Kim and Barnaby climbed down the ladder to the blast-blackened sand. The sergeant of EXTs Provost Marshall who had accompanied them walked with the officers to a hill overlooking the ancient Martian city of Klamagra, which stood on a terrace about five kilometers to the north. The red adobe walls of the city, testimony of the ancient days when Mars had enough water to allow its use for brick-making, blended with the distance to seem a part of the red desert sand.

A cloud of steam and dust appeared between the hill where they stood and the city. Captain Barnaby un-leathered his binoculars and pressed them to the eyepieces of his helmet, and made out a hopping jeep, its top enclosed in plastic and a trio of supercharger coils poking through the sides of the hood. Clouds of steam followed the jeep as its exhaust streamed out into the chilly air.

In a moment the jeep spun up the hill and ground to a halt. There was a pause as the men inside the jeep fitted their helmets on their shoulders, checked their air-pumps, and drew on their gauntlets. Then the plastic bubble lifted back, a sergeant jumped out from under the steering wheel and saluted, and a Colonel, EXTs Intelligence, walked up to Captain Barnaby and Lieutenant Kim. "Gentlemen," he said, "I'm Colonel Lee Montgomery, Commanding Officer, Third Sector. It is my unpleasant duty to turn you over to the Chief Technician of the Martian Judging Authority, who is Rhinklav'n, here."

At these words a tall Martian unfolded himself from the back seat of the jeep. He climbed out and bowed before Captain Barnaby. "I am Rhinklav'n,

Captain." The thick fur nose-flaps, looking like ear-muffs pulled across his muzzle, muffled Rhinklav'n's high-pitched voice so that it gave the effect of coming from the bottom of a rain barrel. "You are to accompany me to Klamugra to be judged by the Machine, of which I am the Honored First Technician."

Barnaby and Kim bowed slightly to acknowledge Rhinklav'n, then crawled into the back seat of the jeep, next to Colonel Montgomery.

Rhinklav'n and the sergeant sat up front. The sergeant pushed a button on the instrument panel, and the plastic top of the jeep dropped down to cover them. As the engine started, the jeep's air-pump drew in air until the atmosphere was thick enough for human lungs. The Martian squirmed uncomfortably in the heavy air while his human companions threw off their helmets. Lieutenant Kim gratefully drew a deep breath of air, and regretted it immediately. What with the million-year water shortage the Martians had lost even the word for bath. Besides, the most popular article of Martian cuisine is a bulb strikingly similar to the terrestrial garlic plant. Captain Barnaby turned to Kim. "Mars has a distinguished atmosphere, hasn't it?" He spoke in English, rather than in the Esperanto lingua-franca of space.

"Indeed it has," Kim agreed. "What was old fuzz-face up there talking about when he spoke of 'the Machine,' Colonel?"

"The law of Mars is the most rigidly systematized in the solar system," Colonel Montgomery replied. "Several millions of years ago, a bright Martie got the idea that it was unwise to trust mortal judges with a problem so important as the sentencing of criminals. So he called in a lot of mechanics—ancient Mars had some pretty fair engineers, though they never discovered electricity—and had them build a judging-machine. Since the climate is right and the machine was built of a stainless steel, it's still here and still being used. It's an enormous thing; spreads over half-an-acre in a big amphitheatre in the center of town. It's an analogue computer, rather clumsy by terrestrial standards, but nevertheless well-built. You know the principles of analogue calculators. Instead of working with coded, position-valued impulses, like the electronic astrogator on our rockets, the mechanical machines solve problems by making use of the physical analogies between cogs and gears and differentials."

"Do you mean that we're going to be punished or set free by a bunch of clockwork, colonel?" Kim asked.

"In a way, yes. The Machine is a most impersonal judge. That fact won't help you, though. Martian legal code is strict about killing, there being some thirty-odd degrees of murder, ranging in seriousness from a 'simple homicide to secure a mate,' the punishment for which is death by dehydration, most often; to 'killing to secure for oneself material benefits,' for which there exist more subtle forms of death by torture."

"Like getting a small-head-size in a vise?" Captain Barnaby grunted.

"That's the usual punishment for murder in the seventeenth degree, where the crime is usually 'killing for spiritual advancement.' You see, each crime is given special study by the Machine. A great many factors are fed in, collated with certain constants within the Machine, processed through several dozen stages, and finally combined into a single number, which represents the punishment called for. By the way," the colonel studied the back of Rhinklav'n's head, "no consideration of the truthfulness of the 'defendant' is entered into the Machine. It is presumed that should a man say that he did not commit a crime, he didn't; if he did, he'd admit it. Martians have a peculiar character defect that prevents them from lying."

"A defect from which we humans are fortunately free," Kim grinned.

"That's no out," Colonel Montgomery countered. "They have witnesses who saw Klaggchallak fry. Besides, we prefer to have the mass of Marties ignorant of the average earthling's penchant for prevarication. It saves the Service a lot of money not to have to prove anything it tells our hairy hosts out here."

The jeep hit the first of the series of low terraces which set the city of Klamugra up from the surrounding desert plains, and the little car bounced high off the sand. Colonel Montgomery looked startled, as though he'd just remembered something. "You know, in my ethnological fervor I didn't realize what you two men are in for. Cosmos! I'm practically delivering you up as human sacrifices!"

"We came to that conclusion five days ago, colonel," Lieutenant Kim dryly observed.

"I can see what the Fleet Commander meant when he said that he was giving me 'a most unpleasant assignment.' Hell, I don't think the Machine is able to give a judgment of 'not guilty'." Colonel Montgomery gazed toward the city they were approaching. "We've got to turn you in. We can't risk a blowup with the Martian Grand Council. There are rumors that ..." the colonel glanced again with suspicion at the back of Rhinklav'n's hairy neck, as though suspecting that the Martian might be able to puzzle out the meaning of their conversation, though it was in English. "There are rumors that the -artiansMay have an agent among the -ussiansRay. We can't risk having the borsch-eaters more popular out here than the Western Powers." The jeep bounded up the last of the terraces and through an opening in the city wall. The adobe buildings raced past, and with a final bound the jeep came to the edge of the huge, circular bowl which held the Machine.

"There's your judge," Colonel Montgomery said, speaking in Esperanto again. "I haven't much hope to offer you. For one thing, you're the first humans ever to be judged by the Machine."

The men picked up their helmets and air-pumps and adjusted them on their shoulders. Rhinklav'n drew his furry nostril-flaps down into place against the sudden change in pressure. The plastic top of the jeep flew back on its springs and the men climbed out, stretching their cramped muscles. The radiophones in the helmets buzzed, and the colonel gave Captain Barnaby a last word. "I want to impress you with the fact that the Service cannot protect you, from this moment onward. If you escape being killed it must be on your own merits. And don't start shooting Marties—won't do you a bit of good. There's a lot at stake for Earth here. Good luck, men!" Colonel Montgomery saluted, and he and his sergeant jumped back in the jeep, slammed the top down, and whirled away.

Rhinklav'n turned to the two EXTS officers. "Gentlemen, I've assigned you quarters here, near the Machine. Will you follow me?" Kim and Barnaby followed the Martian a short distance from the edge of the amphitheater to a lone adobe building, one story high and about ten meters square. "Here are your quarters, where you'll stay tonight. Your judging is set for tomorrow morning."

Captain Barnaby glanced into the building and was surprised to see that it closed with an airlock, had terrestrial canned foods on neat shelves, and had regular Service cots in place of the rough *mal*-leather mats that the Martians slept on. "It was good of you to go to all this trouble just for myself and Lieutenant Kim, Rhinklav'n," the Captain said.

The Martian paused at the door. "It's not just for you, Captain. Five other terrestrials have committed crimes of various proportions within the last few weeks. They will also be tried here, after your case is disposed of." Rhinklav'n left, considerately closing the airlock door and starting the pump on his way out.

Lieutenant Kim took a can of "B" ration beans down from the shelf and thoughtfully began to open it with his Service knife. "Captain," he said, "this sort of thing could drive our Service from Mars. If the Marties consider it their right to judge every Earthling who runs a jeep into a farmer's *mal* or lands half a meter too near one of their cities, we won't have a man on the planet in a couple of years."

"Kim, we're precedents."

"What do you mean, Yankee?"

"If the Marties succeed in convicting us of murder in some unheard-of degree by using that overgrown Erector Set of theirs, we'll be only the first two of a long string of EXTServicemen to be executed under Mars law. We can't let them do it." Captain Barnaby paused a moment to pour himself out a plateful of beans. "Kim, what was that process you used to rely on back in EXTS Academy in Denver? The one that gave you the right answers after you found that your first solutions to our astrogation problems were a few hundred thousand kilometers off?"

Kim stopped chewing for a moment in surprise. "You mean that you got through the Academy without using the 'finagle factor'? No wonder you made captain so soon. It's simple: I'd look up the right answer in the Service charts, find by what factor my solution was off, and introduce that factor into my next calculation, making it inconspicuous under a lot of mathematical camouflage. Don't bawl me out about it, Barny; I just couldn't see letting my extracurricular activities suffer for my schoolwork."

"Yes, you did a lot of your studying at the Denver Dive. No matter, little man. Eat hearty and get some sleep." Barnaby stirred his beans thoughtfully. "We've got a big day ahead of us tomorrow."

Early the next morning a subordinate technician of the Machine hammered on the airlock. The two terrestrials pulled on their heavy jackets, fur boots, and gauntlets, started the little air-pumps on their shoulders, and opened the lock. "The honored First Technician of the Machine invites your presence at your trial, which is to begin very soon," the Martian said, speaking halting Esperanto. Kim and Barnaby followed him to the edge of the Machine bowl. There had been several changes made during the night. An elevated platform had been set up, identical to the one used in the bloody execution they'd witnessed. About twenty Martians were clustered around the Machine, some of them making last-minute adjustments in the mechanism; others, evidently sightseers, gazing curiously at the two principals in the trial.

Rhinklav'n was waiting, his nose-flaps drawn over his nostrils to keep the cold morning air from cutting into his lungs. "I am pleased that you come," he said. "The Machine is fully assembled for your problem." He pointed down toward the Machine, a vast cluster of separate stages connected by rods. "On the far right, in that small building, is the power source of the Machine, a mercury-turbine engine. We can't spare the water to make steam, you know. The first stage contains the Martian actuarial tables, the second has the actuarial system for determining the probable life-spans of you two Earthlings. That's without taking into consideration the probability that you two will be executed as a result of the judgement of the Machine."

Lieutenant Kim nodded. "Most ingenious. But I'm afraid that there's a factor that you've omitted."

"We've made no factual error, Lieutenant," Rhinklav'n insisted. "The value of the sage Klaggchallak is represented there—" he pointed to the fourth stage, "and your social value to the people of Mars is here represented." Rhinklav'n waved one mitten-like hand toward the fifth stage. "If you'll examine that stage, you'll observe that your value is negative: the shaft

representing it revolves in a direction opposite to that of the others. Yes, you'll surely be executed."

Captain Barnaby nodded, as though the reiteration of the probability of his early demise troubled him less than the philosophical question he'd stumbled across. "Still, as my subordinate officer has said, there's a factor which you seem to have omitted. In the terminology of terrestrial psychomentering, this quantity is called—what did you say it was, Lieutenant Kim?"

"The 'finagle factor,' sir."

"Really?" Rhinklav'n asked, the light of scientific inquiry in his eyes. "I thought I'd taken all the variables of Earthling physiology and psychology into consideration when I set up the plans for the trial. What are the mathematics of this 'finagle factor'?"

Captain Barnaby put one foot up on a connecting shaft, as though he were in the Denver Dive, discussing the relative merits of two video dancers. "As you've doubtless noticed in your extensive study of the Terrestrial mind, Sir Honored First Technician of the Machine, we of Earth place almost equal value on a man's intelligence and on his financial standing as criteria of his worth."

"Yes," Rhinklav'n mused, "I noticed your preoccupation with both intelligence (a minor mental quality, by the way; far inferior to spiritual insight or time-sense) and with the individual's possession of Western Credits."

"As I said," Captain Barnaby continued, "there exists a precise formula, developed by the...."

"By the Noyoudont Dentifrice Laboratories," Kim supplied.

"Yes; their laboratories developed the mathematics of the finagle factor. Briefly, it is this: the square root of the product of Intelligence Quotient over one hundred times the number of credits the individual has

outstanding. Or, written algebraically:" Barnaby knelt down and traced in the sand with his gloved index finger:

$\sqrt{(\text{IQ}/100 \times \text{money in the bank})}$

"Quite a simple equation, easily represented on the Machine," Rhinklav'n observed. He called a subordinate technician to his side and spoke to him in the clicking polysyllables of the Martian language. Turning again to Captain Barnaby, he asked, "And what are the values of 'IQ' and 'Money in the Bank' for you and Lieutenant Kim?"

"Our combined IQs total about 243. How many credits do you own, Lieutenant Kim?"

"Hell, sir; I've got more debts than credits."

"Figure up your debts then, Lieutenant."

Kim raised his right gauntlet, drew a pad of paper and a pencil from a pocket at the back of his hand, and scribbled rapidly. "If we get out of this, Captain, I'll owe about 1046 credits. Subtracting pay due for the last semi-annual period, I owe 437 credits."

"And I have debts totaling 600 credits," Captain Barnaby said thoughtfully. He turned to Rhinklav'n. "The debts of myself and Lieutenant Kim, Sir Honored First Technician of the Machine, total 1037 Western Credits. Being debt, that's a negative number, of course."

"Of course, Captain," Rhinklav'n agreed. "The Machine can handle any sort of number, even a negative number. You noticed that your social value to Mars was easily represented as a minus-number." Rhinklav'n talked rapidly to his assistant and handed him the values of the finagle factor, rewritten in Martian ideographs. He faced Captain Barnaby again. "It will take us about an hour to enter this new factor into the Machine," he said. "You'll not mind waiting?"

"No, not at all," Barnaby murmured. He and Kim leaned against the inside wall of the amphitheater, watching the Martian technicians hurry about; they removed gears and replaced them with gears of another ratio; they connected a stage consisting of eccentric cams strung on shafts; and they installed a mass of machinery at the sixth stage, where the operation of extracting square root was to take place. Kim, comparing the heavy gears

and levers of the Machine with the compact tubes of his electronic astrogator, remarked, "It's like using a trip hammer to crack a walnut."

After a few minutes of watching, Kim and Barnaby became conscious of an intruder within their helmets, a most unpleasant odor. They glanced up to the edge of the bowl. The Martian sightseers were sitting up there, dangling their legs above the Machine and utilizing the pause in the proceedings to eat their picnic lunches. They were busily unwrapping bundles of food from the *mal*-skin pouches hanging by their sides and eating as they watched the technicians work over the Machine.

One of the tourists, judging from his height a young male, threw a small parcel toward Kim. The lieutenant picked it up and unwrapped it. The stench of Martian garlic became unbearable as Kim stared at the unidentifiable tidbit of meat the Martian had thrown him; the air-pump on his shoulder drew the redolence into his helmet in such quantities that Kim's eyes burned. He gestured to show that, while his every instinct demanded that he eat the delicious morsel, he couldn't take his helmet off to do so. With an elaborate pantomiming of sorrow, Kim pitched the gift back up to the Martian boy.

A few adjustments later the technicians filed up from the Machine pit. Rhinklav'n walked over to the two EXTS officers. "If you gentlemen will accompany me, we'll begin the trial at once."

Kim and Barnaby walked together up the steps that led from the Machine, then turned and looked down at the dozens of stages of complex machinery, into which memory and intelligence of a sort had been built. Rhinklav'n pointed toward the fifth and sixth stages. "It is there that the combined finagle factors of you men will be calculated. The fifth stage is quite simple; it will perform the necessary division and multiplication. The sixth stage will extract the square root of the product derived by the fifth. The next six stages of machinery contain the variables of terrestrial behavior, which I and my colleagues calculated from Earth texts. The other stages on the field, fifty-three of them, will collate the results of the calculations of the first twelve stages with our legal code and determine punishment. The final

product will appear at the sixty-seventh stage, represented as the speed of rotation of a single shaft. The revolutions-per-time-interval are decoded by a simple formula to determine the punishment to be levied upon you. Doubtless, it will be some unpleasant form of death."

Kim muttered that he wished that Martians had a bit more tact.

Rhinklav'n waved a hairy arm toward his assistant who had remained below in the Machine pit; and that Martian ran to the power house to start the mercury-turbine engine that ran the Machine. With a whistling that set the thin atmosphere trembling for miles around, the turbine began to turn.

The sightseers on the edge of the amphitheater wrapped up the scraps of their lunches, replaced them in their *mal*-skin picnic hampers, and stood up to watch the Machine. Kim and Barnaby paced up and down along the edge of the bowl, looking down upon the mechanical cerebation being performed by the huge Machine. With a smooth transfer of power from one stage to the next, the first problem—the probable duration of Klaggchallak's life when it had been interrupted by the jets of the *Vulcan*—was solved, and the mechanism of the second stage began to revolve.

"They're seeing how long we can be expected to live, now," Captain Barnaby commented.

That problem fled through a mass of gears and cams; and the partial solution, the sum of the two earthling's life expectancies divided by that of the priest Klaggchallak, ran across a shaft to the third stage, which would determine the old priest's value to the society of Mars. On into the fourth stage the problem flowed, to combine all previous factors with the earthlings' social value to Mars, a negative number.

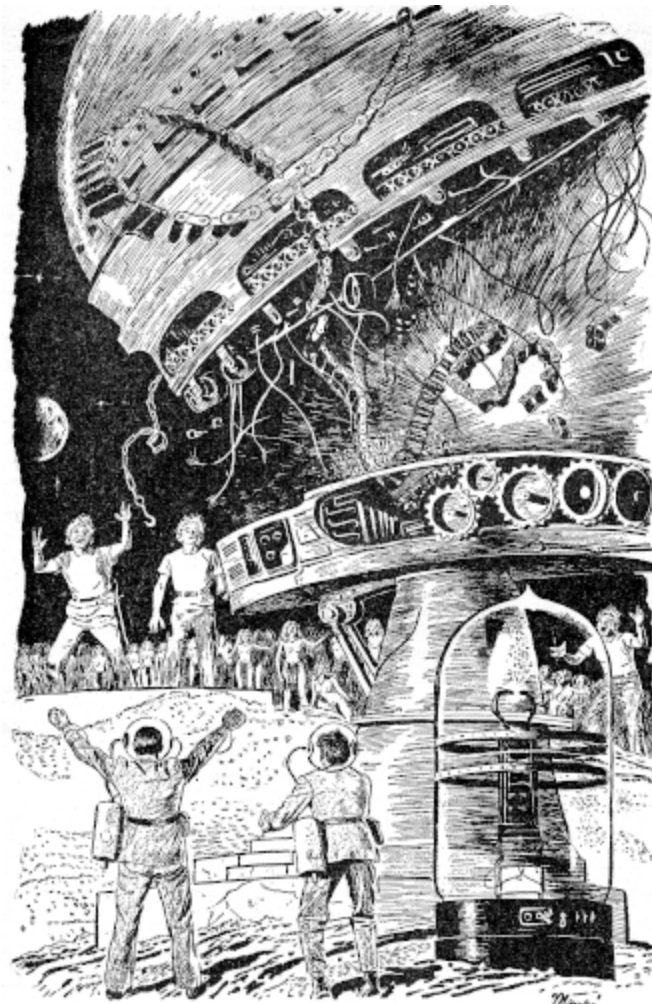
In a few moments the problem had progressed to the fifth stage of the Machine, where the first steps of the 'finagle factor' were solved. The product, a negative number as could be seen by the reversed rotation of the main shaft, bowed into the sixth stage, which was to extract square root.

The turbine howled protest as it was forced to overcome the inertia of the sixth stage; but a governor at the input stage held the shaft-speed constant.

The seventh stage, all ready for the problem when it should appear from the sixth, held all the computations of the first four stages in its smoothly-turning entrails. The initial portion of the sixth stage began to move slowly.

There was a sudden, grating noise as the feed-in gear of the fifth stage came in contact with a solution gear of the sixth which refused to move. The whine of the mercury-turbine engine was shaking the ground beneath the two officers' feet now.

As the Martian technicians and picnickers looked on in amazement, the shaft between the fifth stage and the sixth began to twist like a stick of moist putty. The sixth stage strained and shuddered, then followed the twisting shaft over, tearing its moorings from the ground and smashing upside-down. The seventh stage entered into the chaos, ripping out anchors of steel-in-concrete and slamming onto its side. In a moment all the machines in the bowl were muttering and straining against the earth. Rhinklav'n ran to the stairway that led down into the pit. In the adobe powerhouse the mercury engine was whirling at twenty times its optimum rate, tearing the atmosphere with the sound of its screaming power. There was the rattle of shrapnel exploding within the walls of the powerhouse as the turbine threw off the restraintment of its governor. The whole field within the bowl was a mass of twitching clockwork, shaken by the final stormings of the suicidal turbine. Bars of shining steel twisted and snapped, gear teeth flew singing through the thin air. The final chain of stages tore itself loose from anchoring and crashed to its side. There was a final roar of defiance from the turbine, and the powerhouse walls dissolved before an out-rushing blast of superheated mercury. Kim and Barnaby threw themselves to the ground as the din increased for a moment, and the Martian sightseers sought refuge behind nearby buildings. Suddenly, the Machine was silent, except for the tinkle of scraps of metal falling to the cement.



Bars of shining steel twisted and snapped, gear teeth flew singing through the thin air....

"Looks as though we were too much for judge, jury, and D.A.," Kim murmured into his radiophone. Barnaby nodded, then cautiously climbed to his feet.

Rhinklav'n climbed back up the stairway to the brink of the amphitheater-become-junkyard. He shoved his way through the questioning crowd of Martian sightseers without a word. "Looks like he's going to cry," Lieutenant Kim commented into his radiophone. True, Rhinklav'n's nose-flaps were hanging limply down below his chin, a sure sign of great emotion in a Martian.

Rhinklav'n faced Captain Barnaby wordlessly for a moment. "You may leave now," he said at last. The Martian turned his back on the captain to look down again on the wreck that had been his beloved Machine.

The two EXTS officers wandered about Klamugra, the cynosure of all Martian eyes, though no one tried to stop them or ask them questions. Lieutenant Kim finally spotted a radio tower jutting up above the red adobe buildings. Hurrying in the direction of the tower, Kim and Barnaby found the Klamugra headquarters of the Extraterrestrial Service.

Colonel Montgomery jumped to his feet as they came in, a look of bald disbelief on his face. "Man, I'm glad to see you two! I was about to storm out like a knight in shiny armor and save you from the Marties." He waved his hand toward the helmet and rifle lying on his typewriter table—"If I'd gotten there too late, I'd have ruptured interplanetary friendship for sure!"—and indicated a decanter on his desk. "Have some: that's Edinbrough scotch, not Los Angeles moon-dew. Tell me why I happen to be talking to you now instead of making up a couple of packages for your next-of-kin."

"We wrecked their damn Machine," Kim said happily, dropping his helmet and gauntlets to the floor and measuring out several fingers of the colonel's scotch into his ration can.

"To be a bit more accurate," Captain Barnaby corrected, "we drove the Machine insane." He poured himself a stiff shot of scotch and downed it with appreciation.

"Our personalities are so complex that the Machine blew up all over the landscape when it tried to understand them," Kim said. He dragged a chair out from behind the typewriter table and sat down, carefully balancing the ration can.

"It's rather as though we should set our electronic astrogator to work on a problem with three variables in five dimensions, rather than in four," Captain Barnaby explained. "As you told us, the Machine was a mechanical-analogus calculator. It can multiply, divide, add, square and

cube, and extract roots. It performs these operations by coding numbers into mechanical relationships."

"Just a big adding machine," Kim commented irreverently.

"And our 'finagle factor' was too much for a mechanical system." Captain Barnaby briefly explained to the colonel how he and Kim had induced Rhinklav'n to add their invented factor to the Machine's setup. "You see, the finagle factor resolved itself into the square root of a negative number. An electronic calculator, like our astrogator, could extract the root of a minus-number: 'imaginary' numbers of this sort are implicit in its circuit. The Martian Machine out there couldn't do this though. Since there is no mechanical analogue for an imaginary number, the Machine tried to extract the square root of our finagle factor in the same manner in which it would attempt to extract the root of a real number."

Kim drained, his ration can neatly and remarked, "The Machine couldn't do what it had to. All the power of the turbine was thrown into the root extracting system, which wouldn't revolve. So the Machine went nuts, pardon me, sir, and blew its top. Wrecked the power source and all sixty-seven stages. With the square root of minus one, we busted up a Machine half a million years old."

"What now?" Colonel Montgomery asked, rhetorically.

Captain Barnaby studied the bottom of his ration can a moment. "Well, sir, Rhinklav'n was more puzzled, than angered. He wanted to judge humans not out of malice, but from a genuine scientific curiosity. He wanted to see how the Machine would act with an alien problem. His Machine is too badly broken-up ever to repair. He'll have to find another method of judging criminals, first of all. Martian society is founded on strict law."

"Just a moment." The colonel got up from his desk and went down the hall to a door marked "Judge Advocate General's Department, EXTS." He returned with a heavy book, bound between khaki-board covers. "We'll give this book to Rhinklav'n, and you gentlemen may return to the Denver Joint."

"Dive, sir," Kim corrected.

"Yes, Lieutenant." Colonel Montgomery handed the big book to Captain Barnaby. "Take this to Rhinklav'n before you leave, Captain."

Barnaby turned to the title page and read in Esperanto, "Blackstone. *On the Study of Law.*"

*** END OF THE PROJECT GUTENBERG EBOOK MACHINE OF
KLAMUGRA ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, "Information about donations to the Project Gutenberg Literary Archive Foundation."
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.